From SNNs to Silicon: Compiling FPGA Bitstreams from Spiking Neural Networks

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DTU[2]
TU Delft[3]
ETH zürich[4]

Up-to-date Slides

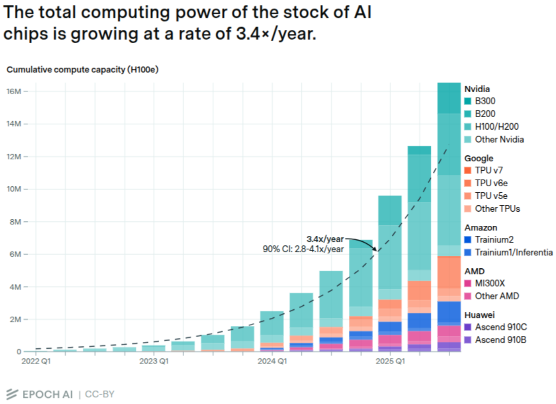
Note: You can find up-to-date slides using the following link

https://neuralfpga.github.io/26iscas/

Why this tutorial?

- **Neuromorphic computing** is a potential solution the scalability issues of neural network-based AI.
- The current *bottleneck* for large-scale neuromorphic computing is **co-design** [a].
- We would like to demonstrate our framework **NIR2FPGA**, an *end-to-end design flow* for
  1. Simulating,
  2. Quantizing,
  3. and Compiling

  RTL from neuromorphic algorithms.

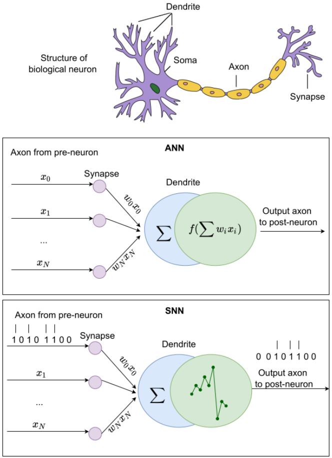


Source: Sevilla et al., "Compute Trends Across Three Eras of Machine Learning," Epoch AI, 2022. CC BY 4.0.

[a] Dhireesha Kudithipudi et al. 'Neuromorphic computing at scale'. In: *Nature* 637.8047 (2025), pp. 801–812

What is Neuromorphic Computing?

- **Neuromorphic Computing**: Exploring parallels between biological neurons and neural computation.



Biological neuron vs Artificial Neural Network (ANN) vs Spiking Neural Network (SNN) (as in [2])

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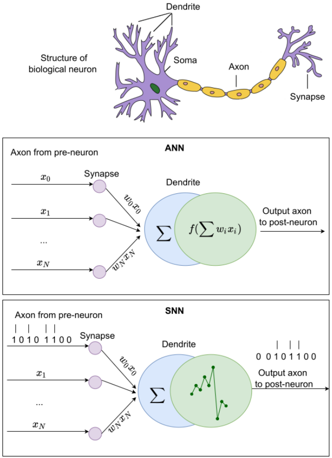

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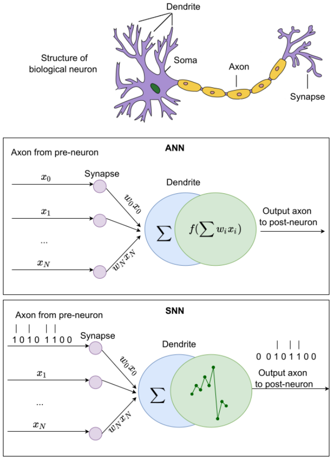

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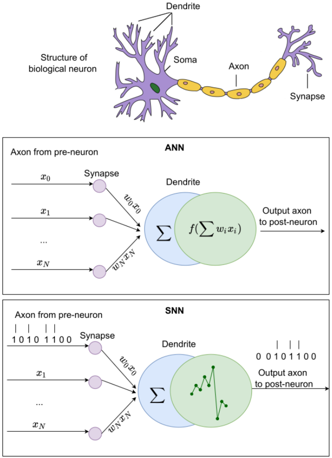

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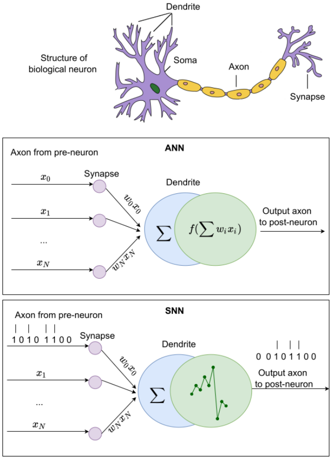

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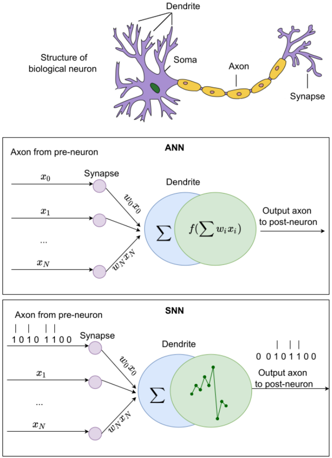
  - *Energy Efficiency*: Event-driven, sparse computation translates to orders-of-magnitude lower power on neuromorphic hardware [2].



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Who are we?

Michail Rontionov
Univ. of Southampton

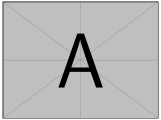
- PhD Student supervised by **David Thomas**
- FPGAs, compilers, languages.
- Tutorial Lead

Jens E. Pedersen
DTU

- Postdoctorate
- Author of Norse and NIR
- Head of Open Neuromorphic

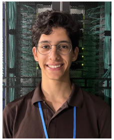
Charlotte Frenkel
TU Delft

- Assistant Professor Delft
- Research Scientist @ Google
- ODIN & ReckOn Online Learning Chips



Francisco Ayala Le Brun
TU Delft

- Bullet 1
- Bullet 2
- Bullet 3



Nassim Beladel
ETH Zurich

- Msc Student EEIT
- Chair of the Young Neuromorphis
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Schedule

Time	Topic	Presenters
08:30	Welcome	Michail Rontionov
08:40	Neuromorphic Engineering and Design Approaches	Charlotte Frenkel
09:00	Neuromorphic FPGA Design Paradigms	Nassim Beladel
09:15	Neuromorphic Intermediate Representation (NIR)	Jens E. Pedersen
09:30	**Hands-on**: Defining, Training & Exporting SNNs	Jens E. Pedersen & Michail Rontionov
10:00	*Coffee Break*	
10:15	NIR2FPGA	Michail Rontionov
10:30	SpinalHDL Introduction	Francisco Ayala Le Brun
10:45	**Hands-on**: • *Livecode*: using NIR2FPGA to develop SNN RTL. • *Exercise*: Defining a LIF neuron in SpinalHDL. • *Exercise*: Evaluating an RTL MNIST SNN.	Michail Rontionov
11:45	Thank you, and goodbye!	All

Hands-on Preparation

- Our hands-on exercises assumes an *account on GitHub.com*, **or** *vscode* with WSL and docker.
- Please go to the link

github.com/mrontio/nir2fpga-iscas

Code → Codespaces → Create codespace on main.